

Snapdragon X2 Elite beats Intel, AMD, Apple chips

From NPU to power efficiency and more

Jayesh Shinde | editor@digit.in

Across every benchmark that matters for on-device AI – UL Procyon, Geekbench AI, TOPS figures, and even platform-level power efficiency – Qualcomm isn't just competitive anymore. It's out in front by margins that reshape expectations for Windows laptops.

What becomes clear when you study Qualcomm's latest data isn't just that the X2 Elite Extreme is fast. It's that the definition of AI performance is shifting toward NPUs designed for sustained, efficient machine learning workloads, not brute-force GPU.

Procyon AI Computer Vision

This slide shifts from theoretical performance to practical benchmarking

using UL Procyon AI Computer Vision, comparing NPU-driven results across modern laptop chips. The Snapdragon X2 Elite Extreme leads decisively with a score of 4151, followed by Apple's M4 at 2121, and Intel's Core Ultra 9 288V at 1866. Intel's Core Ultra 9 285H shows a significantly lower 15,628, while the AMD Ryzen AI 9 HX 370 is marked as not able to run this benchmark.

Qualcomm highlights an "up to 5.7x faster" performance advantage relative to Intel's 285H. This slide underscores the difference between TOPS claims and measured AI workload performance, reinforcing Qualcomm's advantage in real testing environments.

Geekbench AI 1.5 benchmark

This slide presents results from

Geekbench AI 1.5, reflecting broad AI inference workloads across platforms. The Snapdragon X2 Elite Extreme tops the chart with 88,615, well ahead of the Apple M4 at 52,193 and Intel's Core Ultra 9 288V at 48,566. Intel's Core Ultra 9 285H shows a significantly lower 15,628, while the AMD Ryzen AI 9 HX 370 is marked as not able to run this benchmark.

Qualcomm emphasizes an "up to 5.6x faster" advantage relative to Intel's 285H. Detailed test system information is provided for all platforms, ensuring benchmarking transparency and comparability.

Hexagon NPU > NPU + GPU

Qualcomm's latest data positions the Snapdragon X2 Elite Extreme as a clear leader in on-device AI, outpacing both competing NPUs and integrated GPUs by wide margins. In UL Procyon AI Computer Vision tests, the Hexagon NPU posts an industry-high score of 4151, delivering up to 5.7x the performance of Intel's Core Ultra 9 285H NPU. Even against Intel's more capable 288V NPU, Qualcomm maintains a decisive lead. The slide also compares GPU-accelerated AI, where Intel's integrated graphics still fall behind—underscoring that Qualcomm's NPU isn't just faster than rival NPUs, but faster than the GPU fallback path many Windows laptops rely on.

The efficiency numbers sharpen the contrast further. With a Procyon score-per-watt of 200.5, the Snapdragon X2 Elite Extreme delivers up to 3.8x better power efficiency than Intel's NPUs and 3.7x better than Intel's integrated GPUs. Qualcomm's architecture appears tuned not only for maximum throughput but also for sustained performance within tight laptop power envelopes. The overall takeaway is simple: for Windows AI workloads, Qualcomm says its Hexagon NPU currently offers the strongest blend of speed and efficiency, setting a high bar for on-device generative AI and agentic computing workloads. **d**

